

PAGE 1 — FINAL PHASE COVER

The document starts with the whole phase identity before moving to page-by-page engineering detail.

PHASE 14 — AUDIT LOG VIEWER

Read-only governance, timeline intelligence, and Super Admin audit visibility for CMCMIS_SIMPLIFIED.

LOCKED PURPOSE

Phase 14 converts existing audit, identity, and status-history trails into a controlled Audit Log Viewer. It does not create operational records, it does not mutate legacy workflow state, and it does not weaken row-level safety. The module exists so the Super Admin can trace who did what, when, on which entity, and from which client context.

	Locked Value
me	Phase 14 — Audit Log Viewer
rea	Admin / Governance / Audit Intelligence
Doctrines	Strict read-only. All routes are GET. No new operational tables.
	Super Admin only through audit:read-list and audit:export.
e	Uploaded Phase 14 pasted transcript was used as the source of truth.

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PAGE 2 — WHAT PHASE 14 IS

A governance page, not a CRUD page.

DS MISSION STATEMENT

The Audit Log Viewer is not a normal table page. It is the trust layer. It gives CMCMIS a courtroom-grade operational memory where every important state movement and access-control change becomes searchable, explainable, exportable, and linked back to the real module entity.

	Decision / Insight
	Before Phase 14, audit rows existed but were not human-operational. Engineers had data but no governance screen.
	Expose a read-only Super Admin viewer over audit_log, user_role_history, and status-history tables.
nt	No destructive schema work; no writes from the audit module; no mutation controls in UI.
Definition	A Super Admin can filter by time, actor, action, entity, and source, then open a detail drawer with JSON/details/diff.
Position	Audit visibility is sensitive, therefore not granted to LIC, Engineer, Normal User, or View-Only by default.

- Audit is the system memory.
 - Filters are investigation tools, not just UX controls.
 - CSV export is allowed only through audit:export.
 - Deep links connect audit evidence to operational pages.
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PAGE 3 — PHASE 14 STEP MAP

Entire implementation path captured as a senior-engineering sequence.

	Engineering Output
	Introspect audit_log, user_role_history, job_request_status_history, job_card_status_history, and schedule_status_history.
	Seed audit:read-list and audit:export through migration 600; add Audit Log to Admin sidebar.
	Build audit.repo.js as SELECT-only DAL over three source families.
	Build service/controller/routes for list, detail, filters, and CSV export.
	Build FE AuditViewer, AuditDetailDrawer, api/audit.js, and useAuditLog hook.
	Define smoke tests and read-only verification strategy.
	Seal summary and memory update plan.

THINKING MODEL

Phase 14 is built by starting from existing truth tables, not by inventing new audit structures. We first discover available evidence, then normalize it into a single UI language: timestamp, actor, action, entity, summary, notes, diff, and deep link.

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PAGE 4 — STEP 0 DATABASE INTROSPECTION

Real schema discovery before architecture decisions.

STEP 0 FINDING

The transcript began with database introspection. This is correct: audit pages must not be designed from imagination. They must be designed from real columns, indexes, row counts, and table semantics.

	Introspection Insight
audit_log	Primary operational audit table. PK is audit_id, not id. Timestamp is occurred_at.
user_role_history	Identity/access history. Stores user_id and actor_user_id, not employee_id directly.
job_request_status_history	JR lifecycle transition source. PK history_id, entity jr_no.
job_card_status_history	JC lifecycle transition source. PK history_id, entity jc_section_no.
schedule_status_history	Phase 13 schedule transition source. PK id, actor_employee_id, created_at.

Read-only introspection target list

['audit_log','user_role_history','job_request_status_history','job_card_status_history','schedule_status_history']

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PAGE 5 — AUDIT EVIDENCE INVENTORY

The actual tables and row states that power the page.

Source	Observed State
	123 rows observed. Indexes on action+time, actor+time, entity+entity_id+time, and occurred_at.
_history	17 rows observed. Indexed by action, actor_user_id, and user_id with created_at.
y	91 rows observed. Indexed by jr_no+transitioned_at and transitioned_by.
y	11 rows observed. Indexed by jc_section_no+transitioned_at and transitioned_by.
history	7 rows observed. Indexed by schedule_id+created_at.

WHY INDEXES MATTER

The audit page is investigation-heavy. Users will filter by actor, action, entity, and time. The existing indexes already support these paths well enough for an MVP viewer without adding Phase 14 schema changes.

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PAGE 6 — audit_log CANONICALIZATION

How the primary audit table becomes a UI-friendly event row.

	Meaning for Phase 14
id	Primary key. Service must not assume id.
employee_id	Who performed the action. Joined to cmms_emp_mst for display name.
role_code	Role snapshot at action time.
	Stable event verb such as JR_CREATE_SUBMIT, JC_CREATE, USER_ROLE_CHANGED.
type + entity_id	Navigation and filtering identity for the affected record.
ed_at	Canonical timestamp for list ordering and date range filtering.
ress / user_agent / request_id	Client forensic context.
	JSON string or legacy plain text. Parsed safely by service.

REPO ALIASING DOCTRINE

audit.repo.js exposes canonical API fields. Raw columns like audit_id and occurred_at remain inside the repo. The frontend never learns legacy column naming.

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PAGE 7 — IDENTITY & ACCESS HISTORY

user_role_history becomes the second audit source.

	Audit Meaning
	Primary row id for identity-history source.
	Subject user whose role/status changed.
le → to_role	Role transition diff.
tive → to_active	Activation / deactivation diff.
	CHANGE_ROLE, ACTIVATE, DEACTIVATE, CREATE, FORCE_LOGOUT.
er_id	Admin user who performed the change; resolved back to employee_id.
at	Identity-history timestamp.
	Rendered as notes/details in drawer.

IDENTITY SOURCE REASON

Identity and access changes deserve a separate tab because they explain why someone gained or lost system power. Mixing them with JR/JC events would hide the most sensitive governance story.

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PAGE 8 — STATUS TRANSITION UNION

Three lifecycle histories, one audit language.

on Family	Normalized Design
quest	history_id, jr_no, from_status, to_status, transitioned_at, transitioned_by, reason.
l	history_id, jc_section_no, from_status, to_status, transitioned_at, transitioned_by, reason.
	id, schedule_id, from_status, to_status, actor_employee_id, reason, created_at.
strategy	Normalize all three into source_table, id, occurred_at, action, entity_type, entity_id, actor_employee_id, from_status, to_status, r
t	One Status Transitions tab can show JR, JC, and Schedule movement together.

Canonical action synthesis
JR + to_status -> JR_SUBMITTED
JC + to_status -> JC_COMPLETED
SCHEDULE + to_status -> SCHEDULE_COMPLETED

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PAGE 9 — MIGRATION 600 PERMISSION SEAL

New permission codes without disturbing historical grants.

MIGRATION 600 DOCTRINE

Phase 14 uses one permission migration only. No new operational tables, no ALTERs, no data movement. It introduces the permission language needed to gate the Audit Viewer and CSV export.

Decision Element	Decision
read-list	Allows opening the Audit Log Viewer, listing rows, opening detail drawer, and loading filters.
export	Allows CSV export of filtered audit rows.
code	audit_log:read remains untouched for backward compatibility but Phase 14 uses the new audit:* codes.
target	SUPER_ADMIN only.
reason	Audit trails are sensitive and must not be broadly visible by default.

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PAGE 10 — SUPER ADMIN RBAC MATRIX

The Audit Viewer is intentionally narrow and sensitive.

	Phase 14 Access
ADMIN	audit:read-list ✓, audit:export ✓
_CHARGE	No Phase 14 audit permissions by default.
GINEER	No Phase 14 audit permissions by default.
L_USER	No Phase 14 audit permissions by default.
ONLY	No Phase 14 audit permissions by default.

WHY SA-ONLY?

Audit rows expose operational behavior, role changes, entity history, and forensic client fields. This is governance data, not normal workflow data. Expanding access should require a later policy-driven migration, not an assumption.

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PAGE 11 — BACKEND ROUTE MAP

Every endpoint is a read endpoint.

oute	Gate + Purpose
ET /api/v1/audit	audit:read-list — list rows for selected source and filters.
ET /api/v1/audit/filters	audit:read-list — load action/entity dropdown values.
ET /api/v1/audit/export	audit:export — CSV export with row cap.
ET /api/v1/audit/id	audit:read-list — detail row lookup for drawer.

READ-ONLY ROUTE RULE
The route table contains only GET methods. No POST, PATCH, PUT, or DELETE exists in the audit module. This is a structural safety guarantee.

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PAGE 12 — VALIDATION ARCHITECTURE

The outer API perimeter for a read-only module.

or	Role
ySchema	source, from, to, actor, action, entityType, entityId, q, page, page_size.
ramSchema	Positive integer id shared across audit_id/history_id/id sources.
erySchema	source selects table family; subSource can help transition detail lookup.
uerySchema	source only.
uerySchema	Same filters as list but without pagination.
	audit_log, identity, transitions.

STRICT MODE VALUE

Unknown query keys are rejected. This prevents typo-driven silent failure and keeps audit investigation reproducible.

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PAGE 13 — audit.repo.js READ-ONLY DAL

The canonicalization center of the entire phase.

REPO CONTRACT

audit.repo.js is the only audit module file that knows table-specific column names. It converts each physical source into a canonical row. This keeps service and frontend logic stable even when underlying tables differ.

Function	Purpose
auditLog	Reads audit_log with actor/action/entity/time/q filters.
auditLogById	Looks up by audit_id.
entityHistory	Reads user_role_history with user/actor resolution.
entityHistoryById	Looks up identity row by id.
statusTransitions	UNION ALL over JR/JC/Schedule status histories.
statusTransitionById	Finds detail by source hint or default strategy.
act*Filters	Dropdown sources for actions and entity types.

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PAGE 14 — SOURCE 1: ALL ACTIONS

audit_log as the main operational audit source.

	Implementation Logic
	from/to mapped against al.occurred_at with day-boundary expansion.
	al.actor_employee_id exact match.
	al.action exact match.
	entity_type and entity_id exact filters.
	LIKE over action, entity_type, entity_id, and notes.
	occurred_at DESC, audit_id DESC.
	cmms_emp_mst for actor_name display.

Canonical list row idea
audit_id -> id
occurred_at -> occurred_at
notes -> notes_text / notes_json after service parse

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PAGE 15 — SOURCE 2: IDENTITY & ACCESS

user_role_history as account governance evidence.

Logic	Detail
Resolution	actor_user_id joins users, then employee_id joins cmms_emp_mst.
Resolution	user_id joins users, then employee_id/name for affected account.
Filter	urh.action exact match.
	Subject employee_id.
	Role diff, active diff, and target user context.
	/admin/users because individual user detail route is not built in transcript.

WHY THIS IS SEPARATE

Identity history is different from workflow history. It answers who gained power, who lost power, and which admin caused that change.

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PAGE 16 — SOURCE 3: STATUS TRANSITIONS

JR, JC, and Schedule movement in one investigation tab.

on Logic	Phase 14 Decision
	job_request_status_history -> source_table = job_request.
	job_card_status_history -> source_table = job_card.
Source	schedule_status_history -> source_table = schedule.
ynthesis	CONCAT prefix with to_status: JR_SUBMITTED, JC_COMPLETED, SCHEDULE_CANCELLED.
ed	source_table / subSource helps resolve id collisions across unioned tables.
me	Joined after union through cmms_emp_mst.

UNION REASONING

Lifecycle transitions are the most important forensic story. The union turns several physical history tables into one timeline stream while preserving sub_source for precise drilldown.

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PAGE 17 — audit.service.js CANONICAL API

The service converts data into investigation language.

Unit	Responsibility
row, source)	Maps any source row into the frontend event shape.
seNotes	Safely parses JSON notes; preserves raw text for legacy rows.
ummary	Creates human-readable list summaries from action/diff/entity fields.
nkFor	Maps entity_type to known frontend routes.
lit	Chooses repo source, applies pagination, returns applied_filters.
litDetail	Loads precise row and returns canonical detail object.
ers	Caches dropdown options for 15 minutes.
Csv	Builds CSV and row-cap metadata.

SERVICE VALUE

The service is where raw rows become investigation-ready records. It does not query SQL directly; it enriches, normalizes, parses, summarizes, and linkifies.

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PAGE 18 — DETAILS PARSING & DEEP LINKS

How raw audit notes become readable evidence.

Field	Why It Exists
n	Valid JSON notes become pretty printable object data.
t	Plain-string legacy rows remain visible.
	Human-readable short explanation shown in table.
k	Optional FE route to target entity.
back	Unknown entity types safely return null link.
	IP, user agent, request_id displayed when available.

Notes strategy

if notes starts with { or [-> try JSON.parse

else -> keep raw plain text

never crash drawer on malformed notes

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PAGE 19 — FILTERS + CSV EXPORT

Investigation needs both discovery and offline evidence.

/ Filter Feature	Engineering Explanation
Cache	15-minute in-process cache for action/entity dropdown values.
Reader	Timestamp, Source, Action, Entity, Actor, From-To, IP, UA, Request ID, Summary, Notes.
Cap	HARD_ROW_CAP = 5000 for export safety.
Escaping	Quotes and commas escaped before streaming.
Header	X-Export-Rows and optional X-Export-Capped tell the FE what happened.
Reserve audit	Export itself does not create an audit row in this phase to preserve read-only doctrine.

EXPORT DOCTRINE
CSV export is powerful and sensitive, therefore separated from read-list using audit:export. Filtering should be encouraged before export to avoid broad extraction.

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PAGE 20 — FRONTEND AUDIT VIEWER

The main Super Admin investigation screen.

UI Element	Design Meaning
Header	Main /audit page with tabs, filters, URL sync, table, pagination, export, and drawer launch.
Drawer	All Actions, Identity & Access, Status Transitions.
Table	Date From, Date To, Actor, Action, Entity Type, Entity ID, Search.
Badges	Color-coded by prefix: JR, JC, SCHEDULE, PO, SPARE, USER, EQUIPMENT.
Buttons	Timestamp, Actor, Action, Entity Type, Entity ID, Summary, IP, View.
Footer	25 rows/page default with total rows display.

UX INTENT
The page feels like an operations investigation console: filter first, inspect timeline, open drawer only when a row needs deeper explanation.

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PAGE 21 — URL-SYNCED FILTERING

Reproducible investigation views.

Inc Feature	Reason
ams	source, from, to, actor, action, entityType, entityId, q, page.
lity	A filtered investigation can be bookmarked or shared internally.
Behavior	Refreshing browser preserves selected source and filters.
witching	Switching source clears source-specific filters like action/entityType.
ttion	Clears filters without changing source tab.
aram Cleanliness	Default audit_log source can be omitted unless another source is active.

WHY URL SYNC MATTERS

Audit investigations are conversations. A Super Admin should be able to reproduce the exact view that led to a conclusion.

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PAGE 22 — AUDIT DETAIL DRAWER

One row becomes a complete forensic card.

Feature	Implementation Detail
Drawer.js	Right-side drawer with read-only details.
Fetch	Drawer uses useAuditDetail so the row is re-read before display.
	Who, What, Transition, Where, Details.
	Click outside closes drawer.
	Entity ID can open JR, JC, Equipment, Admin Users, Employees, Schedule, Procurement.
	role=dialog and aria-modal=true.

DRAWER ROLE
The table gives breadth; the drawer gives evidence depth. It displays enough context for a governance reviewer without modifying anything.

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PAGE 23 — DIFF + NOTES RENDERING

The drawer turns raw metadata into readable investigation evidence.

	Visible Behavior
gic	from_status -> to_status rendered as colored pills.
f	from_role -> to_role rendered for identity rows.
f	from_active -> to_active displayed for activate/deactivate transitions.
te	Null from value renders as (initial) -> target.
es	Pretty-printed in monospaced block.
d Notes	Never crash; show raw text instead.

Diff rendering rule

if from is null -> (initial) -> to
else -> from -> to
render from in red, to in green

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PAGE 24 — FRONTEND API + HOOKS

Thin wrappers, stable cache keys, and zero mutations.

Layer	Responsibility
.js	fetchAuditList, fetchAuditDetail, fetchAuditFilters, downloadAuditCsv.
List	Paginated list query with keepPreviousData and refetch on focus.
Detail	Enabled only when drawer id exists.
Filters	Caches dropdown values with 15-minute stale time.
IAuditCsv	Blob response + manual anchor download.
rams	Falsy/empty values removed before hitting backend validator.

REACT QUERY DOCTRINE

The audit page is read-only, so the frontend has no mutation hooks. It only reads, filters, refetches, and exports.

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PAGE 25 — SIDEBAR + ROUTE WIRING

The module enters the app through Admin governance navigation.

	Locked Behavior
point	Admin sidebar adds Audit Log entry gated by audit:read-list.
as.js	/audit route imports AuditViewer and uses ProtectedRoute requiredPermission audit:read-list.
ocation	Under Admin panel to match Super Admin governance context.
ety	Without permission, user cannot open the page even if URL is typed manually.
utton	Visible only when hasPermission(audit:export) is true.
Only	View-Only receives no audit permissions from migration 600.

DOUBLE GATE
The sidebar hides the link, and the route itself remains protected. UI hiding is not treated as security.

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PAGE 26 — READ-ONLY SMOKE STRATEGY

How Phase 14 should be verified safely.

Check	Expected Result
Scan	Routes should expose only GET methods.
rb Scan	Repo queries should begin with SELECT only.
ion Smoke	Super Admin should receive 200; non-SA should receive 403.
Smoke	audit:export required; output must be text/csv.
smoke	Dropdown endpoint returns actions/entityTypes.
smoke	Detail route returns canonical row with notes_json/notes_text.
nt Snapshot	audit_log and source table counts should not change after audit page requests.

ZERO-WRITE ASSERTION
A good Phase 14 smoke test should prove not only that the page works, but that using the page cannot create, update, or delete operational data.

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PAGE 27 — VERIFICATION STATUS FROM TRANSCRIPT

What is proven by the pasted evidence, and what is not shown.

ed Verification	Meaning
on Runner	Migration 600 was applied according to migration status output.
Warnings	The general runner still reported old count checks failing: permissions count/users/user_roles expected old values.
etation	Those warnings are not specific Phase 14 failures; they come from old bootstrap assertions after many phases expanded the data
Load	Transcript initiated Node require checks for audit.routes, audit.service, audit.validators, audit.repo.
unt Limit	The supplied transcript ends before showing a final Phase 14 smoke count or module-load output.
entation Position	This DOCX captures completed implementation evidence from the transcript and clearly marks verification gaps.

HONEST SEAL

We can lock the documented design and generated implementation from the pasted transcript. A final green smoke count was not included in the uploaded text, so it is not invented here.

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PAGE 28 — RISKS & MITIGATIONS

Senior engineer review of fragile points.

	Mitigation
Export Risk	CSV export can expose many audit rows. Controlled through audit:export and row cap.
Sensitivity	Role and activation history is sensitive; SA-only access is correct default.
Collision	Transitions source needs sub_source because different tables can reuse history ids.
Liability	notes can be JSON or plain text; safe parser prevents UI crash.
Column Drift	Repo aliasing protects frontend from audit_id/history_id/transitioned_at differences.
Accuracy	Some entity types map to list pages where no detail route exists; acceptable for Phase 14.

ENGINEERING RISK POSTURE

Phase 14 is intentionally conservative: read-only routes, SA-only access, capped export, safe parsing, and no new operational data model.

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PAGE 29 — FILE MAP

Where Phase 14 lives in the production codebase.

	File / Location
	600__phase14_audit_permissions.sql
Module	src/modules/audit/audit.validators.js, audit.repo.js, audit.service.js, audit.controller.js, audit.routes.js
ount	src/server.js mounted /api/v1/audit.
API	src/lib/api/audit.js
Hooks	src/lib/hooks/useAuditLog.js
Page	src/pages/audit/AuditViewer.jsx
	src/pages/audit/AuditDetailDrawer.jsx
n	src/lib/permissions.js Admin sidebar entry.
	src/App.jsx /audit ProtectedRoute.

MODULE SHAPE

Phase 14 follows the sealed module doctrine: validators -> repo -> service -> controller -> routes on the backend, and api -> hook -> page -> drawer on the frontend.

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PAGE 30 — FINAL LOCKED SEAL

The 30th page closes the phase with the audit-governance doctrine.

PHASE 14 — AUDIT LOG VIEWER

FINAL DS SEAL

Phase 14 turns existing CMCMIS operational traces into a Super Admin governance console. It reads from sealed evidence tables, normalizes mixed audit sources, supports investigation filters, exposes a detail drawer, and allows controlled CSV export. It does not mutate workflow state, does not create audit records from viewing, and does not broaden sensitive visibility beyond Super Admin by default.

	Final Statement
d	Phase 14 — Audit Log Viewer
identity	READ-ONLY, SA-only, capped export, canonical service payload.
ctrine	Trace who changed what, when, where, and why across the system.
Value	Run and record a dedicated Phase 14 smoke matrix once the transcript/build output includes final green verification.
t Step	Governance is not decoration. It is the memory of the system.
on	

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